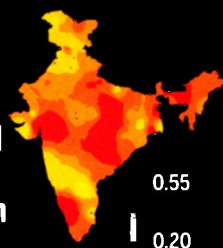


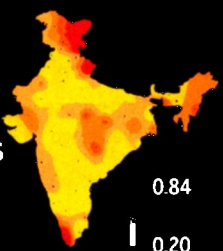
GENE

Recent studies show that some Indians exhibit genetic variations that make them susceptible to specific diseases

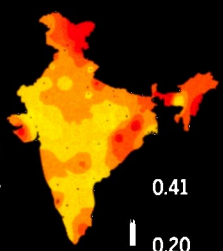
MALARIA
East and Central India populations were found to have a gene that makes them susceptible to cerebral malaria.



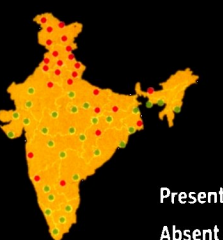
CANCER
Initial studies have found certain populations in the extremes to have higher risk of cancer.



IMMUNE RESPONSE
Indians showed varying immune response to certain drugs holding the key for future treatment.



HEART DISEASES
North Indian vegetarians were more susceptible to heart diseases.



Present
Absent

Red signifies vulnerability

Map not to scale

revelations of how Indians became, well, Indians. Equally significant and directly relevant to the quality of life, for the first time, major genetic studies are underway at a clutch of top national laboratories, which when complete will help us know if any of us run a higher risk of being afflicted with life-threatening diseases such as cancer, diabetes, malaria, cardio-vascular blocks, neurological disorders and asthma. And possibly help in ushering a revolution in healthcare.

Among the more striking results is the finding that Indians are the global melting pot of human diversity containing every possible known genotype. That it is now clear that India was the first stop made by early settlers after the first known human populations migrated out of Africa. (see box on the next page) That the tremendous genetic diversity originated in the sub-continent itself rather than as a result of the so-called 'Aryan invasion' that historians have theorised about. That some regions in Northern India are more susceptible to heart diseases while some populations in Rajasthan and Jammu have a genetic immunity that puts them at lower risk of HIV infections.

The trigger for Indian researchers has been the most spectacular scientific achievement since man walked on the moon: deciphering the book of human life. In 2001, the international Human Genome Project (HGP) came out with a raw transla-

THE GENETIC RISK TO INDIANS OF PARTICULAR DISEASES WAS MORE SPECIFIC TO A REGION RATHER THAN LINGUISTIC OR CASTE AFFINITIES.



BANDEEP SINGH

“India is the first country to do a disease-specific drug-response mapping on large populations. It will give us a risk analysis and also the benefits.”

Samir BRAHMACHARI
Director-General, CSIR